

VR K-Heritage Assembler: Exploring Korean Wooden Architecture Through Virtual Gong-po Assembly

Juwon Lee

Electronics and Telecommunications
Research Institute (ETRI)
Daejeon, Republic of Korea
zacurr@etri.re.kr

Da-un Jung

Electronics and Telecommunications
Research Institute (ETRI)
Daejeon, Republic of Korea
crisia@etri.re.kr

Seungjae Lee*

Electronics and Telecommunications
Research Institute (ETRI)
Daejeon, Republic of Korea
seungjlee@etri.re.kr



Figure 1: VR based *Gong-po* assembling application offers immersive and interactive experiences to understand the structure of bracket set of Korean wooden architecture. Snapshots of proposed application. (a) Selection of building (b) Assembling Interface (c) and (d) screenshot of assembling process.

Abstract

Understanding the structural complexity of *Gong-po*, a key bracket system in traditional Korean wooden architecture, is difficult through conventional resources such as photos and 2D drawings, while direct study is restricted by the fragility of heritage buildings. We present VR K-Heritage Assembler, a virtual reality application that enables users to assemble and explore *Gong-po* structures from 12 Korean national treasure buildings. The system offers immersive interaction and 3D visualization of components, enhancing spatial understanding. A user study with architecture students indicates high satisfaction and strong perceived educational value, suggesting its potential as a learning tool for traditional architecture education.

ACM Reference Format:

Juwon Lee, Da-un Jung, and Seungjae Lee. 2025. VR K-Heritage Assembler: Exploring Korean Wooden Architecture Through Virtual Gong-po Assembly. In *SIGGRAPH Asia 2025 Posters (SA Posters '25)*, December 15–18, 2025, Hong Kong, Hong Kong. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3757374.3771509>

*Corresponding author.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SA Posters '25, Hong Kong, Hong Kong

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2134-2/25/12

<https://doi.org/10.1145/3757374.3771509>

1 Introduction

Gong-po is a traditional wooden bracket system that plays a critical role in Korean architecture by transferring roof loads to columns while forming the aesthetic identity of historical buildings. Despite its cultural and architectural value, *Gong-po* is difficult to study due to fragile structures and limited access.

Conventional learning relies on photos and 2D drawings—abstract representations insufficient for conveying the layered joinery and precise geometry of its structures. Carving wooden replicas offers more realism but demands significant time and cost, making repeated practice impractical. VR enables unlimited, repeatable practice without material/time constraints and has proven effective in assembly-based training, such as car engines [Lan et al. 2023], computer parts [Checa et al. 2023], and timber frame construction [Lucas 2018]. Inspired by this, we present a VR-based learning tool that allows students to assemble *Gong-po* structures interactively, shifting from abstract diagrams to experiential interaction to improve comprehension and accessibility in heritage education.

2 Proposed System

This section introduces VR K-Heritage Assembler, a VR application for assembling and exploring *Gong-po* bracket sets in virtual space.

The system, built in Unity and deployed on the Vive Focus 3 HMD, covers 12 Korean national treasure buildings, letting users select a site and assemble its structure. Accurate 3D models reconstructed from hand-drawn plans, survey reports, and repair reports were converted into CAD models and imported into Unity as part-wise 3D assets for interactive assembly. Fig. 2 shows a model in its

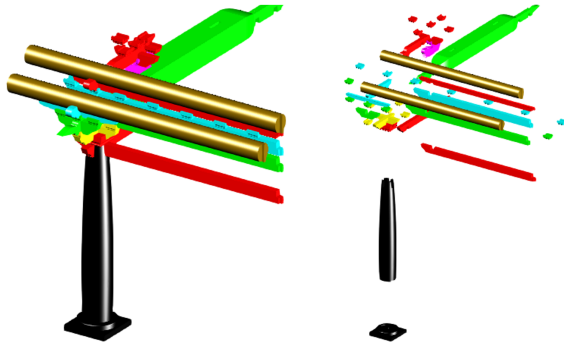


Figure 2: Fully assembled *Gong-po* bracket set (left) and its disassembled components (right).

fully assembled and disassembled forms, highlighting the layered complexity of *Gong-po* joinery the system allows users to explore.

Three difficulty modes accommodate varying levels of VR proficiency and *Gong-po* knowledge: *Easy*, *Normal*, and *Expert*. In the *Easy* mode, all components within each bracket layer (*dan*) are automatically assembled sequentially with minimal user input. The *Normal* mode provides semi-automated assembly: selecting a layer displays component buttons, and clicking one assembles it in the correct position. The *Expert* mode offers full manual assembly—components appear near the controller when selected, and users must drag them precisely into place for successful assembly.

To enhance learning and engagement, the system integrates visual references (photos, drawings), a timer for gamification, and multi-user support with live voice chat. Experts can observe and guide assemblies remotely. With varied difficulty modes, built-in references, and collaborative interaction, the system provides an engaging platform that helps learners of all levels explore *Gong-po* structures more effectively.

3 User Study and Evaluation

We conducted a user study with 11 undergraduate architecture students—7 with prior education in Korean wooden architecture and *Gong-po* structures, and 4 without. This allowed comparison between experienced and novice users.

Each participant received a 10-minute tutorial on system operation, followed by assembly tasks in *Easy* and *Normal* modes. While assembling, they could freely *rotate*, *zoom*, and *move vertically* to observe components from various angles.

After the VR session, participants completed two 5-point Likert-scale surveys (1 = very low, 5 = very high) evaluating satisfaction and perceived educational impact. Fig. 3 summarizes the distribution of responses for both surveys.

User Satisfaction. Participants reported high satisfaction across all categories (Fig. 3a). Overall satisfaction was high, with 100% rating it as High or Very High, and a mean score of 4.55. Interface Design, Ease of Observation, and Immersion each averaged 4.36, with 90–100% positive responses. Appropriateness of Difficulty received a lower mean score (3.91) and was the only item with Low responses, mostly from novices.

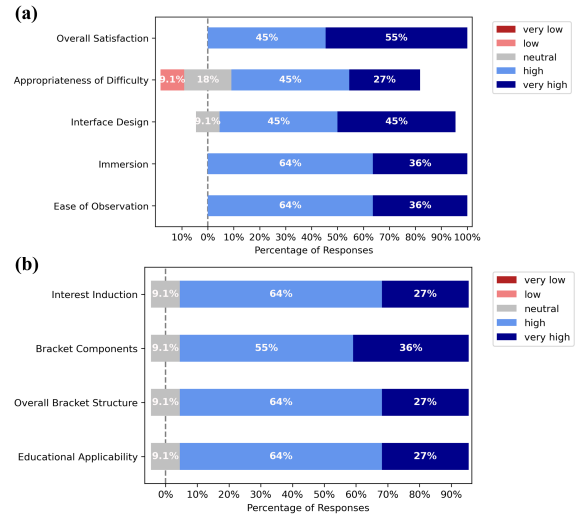


Figure 3: Distribution of the 5-point Likert-scale responses. (a) User Satisfaction (b) Perceived Educational Effectiveness.

Perceived Educational Effectiveness. The second survey assessed the tool's perceived usefulness in traditional architecture education (Fig. 3b). All categories scored at least 4.18, with no negative responses, and 91% were positive, indicating strong consensus on the system's educational value. Understanding of bracket components achieved the highest score (4.27). Notably, novices rated interest induction and educational applicability slightly higher than trained participants, suggesting strong accessibility for learners without prior exposure. (see Supplementary Table 2.)

Overall, results show strong engagement and pedagogical potential. While prior knowledge improves ease of use, even untrained users found clear value in the system as a platform for traditional architectural education.

4 Conclusion and Future Work

This study presented a virtual reality application for assembling and exploring *Gong-po* structures from Korean traditional wooden architecture. The integration of accurate 3D models with adjustable difficulty, visual aids, and multi-user features enables an immersive experience that overcomes the limits of conventional teaching.

User studies confirmed both high satisfaction and strong educational value. Trained participants valued usability and difficulty design, while novices rated educational aspects such as interest and applicability higher. The system supports diverse learners, bridging expert knowledge with public accessibility.

Future work includes expanding participants and adding objective metrics (e.g., pre/post-tests, assembly accuracy) to better quantify learning outcomes.

Acknowledgments

This work was supported by 2024 Cultural Heritage Smart Preservation & Utilization R&D Program of Cultural Heritage Service, National Research Institute of Cultural Heritage (Project Name: Development of AI-based analysis and design technology of traditional

wooden architecture, Project No.: RS-2024-00396158, Contribution Rate:100%)

References

- David Checa, Ines Miguel-Alonso, and Andres Bustillo. 2023. Immersive virtual-reality computer-assembly serious game to enhance autonomous learning. *Virtual Reality* 27, 4 (2023), 3301–3318.
- Gongjin Lan, Qiangqiang Lai, Bing Bai, Zirui Zhao, and Qi Hao. 2023. A virtual reality training system for automotive engines assembly and disassembly. *IEEE Transactions on Learning Technologies* 17 (2023), 754–764.
- Jason David Lucas. 2018. Immersive VR in the construction classroom to increase student understanding of sequence, assembly, and space of wood frame construction. *J. Inf. Technol. Constr.* 23, November 2017 (2018), 179–194.